Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Full-stack Software Engineer with 2+ years delivering **TypeScript**, **React**, Python, and Go services on **AWS** with **Postgres**, NoSQL, containers, serverless, and infrastructure as code, from design through production support.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Architected a multi-tenant platform with normalized **PostgreSQL** and Azure SQL schemas and row-level security for 500+ users, cutting dashboard latency from 1.2s to 350ms through query and indexing optimization.
- Restructured fault-tolerant async job orchestration with FastAPI workers, explicit state transitions, and failure isolation, tripling throughput across 1K+ jobs and sustaining 99%+ **production reliability** for 5K+ monthly integrations.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Delivered a full-stack **React** and Python grading platform from design through production, serving 150+ instructors and 10K+ daily requests on AWS with stable API latency during peak load.
- Built concurrent batch processing with secure PDF, DOCX, and ZIP ingestion through **RESTful** APIs and AWS RDS, handling 1K+ files per batch and cutting feedback turnaround 60% via CI/CD releases.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- **Led a team** redesigning ETL workflows on AWS Glue and S3, boosting pipeline efficiency 68%, cutting storage costs 15%, and reducing technical debt 20% through object-oriented Singleton and Factory designs.
- Troubleshot large-scale API integrations, improving response times 30%, and automated nightly Jenkins **CI/CD** with 20+ unit and integration test suites across three operating systems in Agile sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight microservices with **TypeScript**, Node.js, React, and MongoDB for authentication, resources, billing, leads, and notifications, synchronizing 2M+ records across locations through REST APIs and real-time WebSockets.
- Provisioned containerized services on AWS ECS with **AWS CDK** infrastructure as code and shipped Docker builds through Jenkins CI/CD with staged approvals, reaching 95%+ build validation across releases.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built **Go** REST APIs for bookings, time-slot reservations, pricing, coupons, and payments with Firebase real-time sync, sustaining sub-200ms latency and 40% faster confirmations across 50K+ booking records.
- Designed the end-to-end architecture and 12+ integration points with the client, using **transactional locking** to prevent double-booked slots and idempotent payment webhooks that cut manual refund handling 70%.

Skills

Languages & Frontend: TypeScript, Go, Python, JavaScript, SQL, Java, HTML, CSS, React, Next.js, TailwindCSS
APIs & Data: RESTful API Design, Node.js, FastAPI, Flask, WebSockets, PostgreSQL, MongoDB, DynamoDB, Firebase
Cloud & Infrastructure: AWS (Lambda, ECS/Fargate, S3, RDS), AWS CDK, Infrastructure as Code, Serverless, Docker, Kubernetes
Delivery & Reliability: CI/CD, GitHub Actions, Jenkins, Sentry, PyTest, Unit Testing, Integration Testing, Troubleshooting
Practices: System Design, Microservices, Code Reviews, Mentoring, Agile/Scrum, Cross-functional Collaboration

Projects

Community Event Registration Platform | *Go, AWS Lambda, DynamoDB, Postgres, React, TypeScript, AWS CDK*

- Designed a serverless event-registration platform with **Go** Lambda REST APIs, DynamoDB conditional writes enforcing venue capacity, Postgres tournament standings, a React/TypeScript organizer console, and **AWS CDK** infrastructure as code.

FinNexus Live | *Next.js, TypeScript, MongoDB, WebSockets, Kubernetes, TailwindCSS*

- Built a **TypeScript** and Next.js fintech platform with MongoDB, Plaid bank linking, and WebSocket real-time updates, deployed as containers on Kubernetes with CI/CD for 50K+ users and 1M+ transactions.

Linguistix | *Python, Docker, AWS ECS/Fargate, CodePipeline, CI/CD*

- Developed and deployed a containerized Python REST data processing API on serverless **AWS ECS/Fargate** with automated CodePipeline CI/CD releases and validation checks, achieving 87% accuracy and production-grade reliability.